From Point Forecasts to Distributions: Adapting TimesFM to Synth

SYNTH
www.synthdata.co

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TL;DR

Synth (Bittensor Subnet 50) is a forecasting competition. Each miner submits not a single prediction but a *spread* of possible future price paths for a set of assets, and is scored on how well that spread matched reality. Being right on average is not enough—you are graded on your uncertainty.

We took a strong off-the-shelf model, **TimesFM 2.5**, and adapted it to Synth as a ladder of seven steps, measuring each. The findings:

- Using only the raw point forecast—replicated across every path—lands near the bottom of the ∼256-miner field.
- **One cheap step does most of the work**: sampling the model's own built-in ranges instead of its single prediction. Mean rank ∼230 → ∼188, error down ∼19%. No training, no GPU.
- Tail completion, drift re-centering and a 15-minute fine-tune each improve the average; per-asset tuning adds nothing measurable. The gains after R2 are small and their CIs overlap. The full ladder reaches mean rank ∼167, with the fine-tune the best and most stable configuration.
- The whole thing cost about three days and under an hour of GPU on a gaming graphics card.
- We do *not* beat our own deployed Chronos-2 miner at the 24-hour horizon (∼167 vs. ∼91). At 1 hour TimesFM is slightly ahead on the point estimates, but the confidence intervals overlap, so the comparison is inconclusive.

The recipe and its cost are the point, not a new champion. Everything is measured across nine (24-hour) and eight (1-hour) market-regime windows with bootstrap confidence intervals.

The setup

For each asset a miner submits an ensemble of possible price paths; the validator scores it with CRPS, which rewards forecasts that are accurate *and* honestly spread. Most foundation models, TimesFM included, output a *point* forecast plus per-step ranges—the wrong shape for a game that scores whole path ensembles.

As a yardstick we use our own live Chronos-2 miner, one of the field's strongest (recently ∼62nd of ∼256 overall, and #1 in its main competition for long stretches). **One caution on numbers:** throughout we rank everything the same strict way—per-prompt error rank against the live field, averaged over the 12 assets and over each window, then across windows. By that strict measure Chronos averages about **91** on the 24-hour horizon (worse than its smoothed leaderboard standing, because it mixes in every asset and day). That 91 is the honest, like-for-like bar; our TimesFM setups start at ∼230.

The ladder (24-hour horizon)

Each rung adds one idea: **R0** raw point forecast (zero spread); **R1** + hand-made Gaussian noise; **R2** sample the model's own deciles (the big win); **R3** extend the tails past the deciles; **R4** remove the model's built-in directional drift; **R5** per-asset tuning (turns out ≈ null); **R6** a 15-minute LoRA fine-tune.

Table 1: 24-hour profile: mean estimated rank and mean weighted CRPS across nine regime windows, with 95% bootstrap confidence intervals. Lower is better; field ∼256. Chronos-2 is our live reference, measured the same strict way.

Arm	mean rank [95% CI]	mean CRPS [95% CI]
Chronos-2 (reference, live)	91 [82, 105]	2264 [1752, 2790]
R0 point	230 [226, 235]	2965 [2283, 3673]
R1 + Gaussian noise	209 [191, 225]	2546 [2098, 3009]
R2 decile sampler	188 [167, 207]	2406 [1853, 2976]
R3 + tail completion	178 [161, 194]	2371 [1839, 2921]
R4 + drift re-center	174 [155, 192]	2356 [1828, 2899]
R5 + per-asset tuning	174 [155, 192]	2356 [1828, 2899]
R6 + LoRA fine-tune	**167 [158, 177]**	**2324 [1805, 2857]**

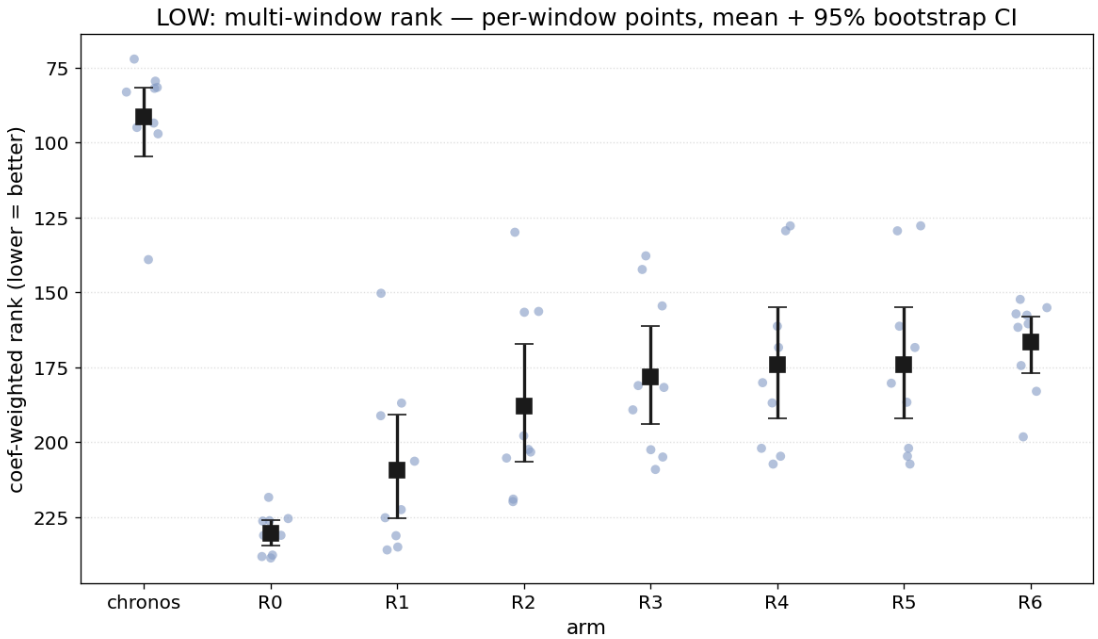

Figure 1: The 24-hour ladder across nine regime windows: per-window points, mean, and 95% bootstrap CI per rung; shaded band = the Chronos-2 reference range. Lower is better.

Two takeaways. (1) The big win is nearly free: use the model's own ranges (R0→R2, −42 mean ranks, −19% error, no training). (2) Most later steps improve the average, while per-asset tuning adds nothing measurable: R6 (the fine-tune) is the best adapted arm (∼167) and the most stable across regimes (its rank stays in a tight band where the plain R2 sampler swings widely). The gains past R2 are small and their CIs overlap.

The 1-hour horizon: TimesFM is competitive with Chronos-2

At Synth's 1-hour crypto horizon the 24-hour levers partly invert—tail completion *hurts* and re-centering is neutral, because the native band is already adequate. But here the fine-tune pays off: the best arm, a fine-tuned decile sampler, averages rank ∼116 across eight windows versus ∼130 for our Chronos-2 miner (and leads on error too). TimesFM is slightly ahead on average, but the confidence intervals are wide and overlapping, so the advantage is too uncertain to call a clear win—and it is helped by the 1-hour profile being Chronos-2's weaker leg.

Why publish it

The gap between "bottom of the field" and "inside the field's main body" costs about three days and under an hour of GPU on a single consumer card—no cluster, no paid data, no proprietary model. The barrier to entry on Synth is know-how, not compute. That is the argument for measuring it publicly.

Caveats: all ranks are estimated from backtests, not live; our setups start from the true price (slightly flattering vs. the live reference); we score 100 paths where the field serves 1000; and the newly-listed asset SPCX could not be backtested on most windows. Exact settings are omitted on purpose. Full version available on request.